STM32F746ZG / Industrial communication board

01 MCU and debug

U1 STM32F746ZGT6

	PE2	PB12	
	PE3	PB13	
	PE4	PB14	
	PE5	PB15	
	PE6	PD8	RS232_TX
V3V3	VBAT	PD9	RS232_RX
	PC13	PD10	
	PC14	PD11	
	PC15	PD12	
	PF0	PD13	
	PF1	VSS	GND
	PF2	VDD	V3V3
	PF3	PD14	
	PF4	PD15	
	PF5	PG2	
GND	VSS	PG3	
V3V3	VDD	PG4	
	PF6	PG5	
	PF7	PG6	
	PF8	PG7	
	PF9	PG8	
	PF10	VSS	GND
HSE	PH0	VDDUSB	V3V3
	PH1	PC6	RS485_TX
NRST	NRST	PC7	RS485_RX
PHY_RESET	PC0	PC8	RS485_DIR
ETH_MDC	PC1	PC9	
	PC2	PA8	
	PC3	PA9	USB_VBUS_SENSE
V3V3	VDD	PA10	
GND	VSSA	PA11	USB_DM
VDDA	VREFP	PA12	USB_DP
VDDA	VDDA	PA13	SWDIO
	PA0	VCAP_2	VCAP2
ETH_REFCLK	PA1	VSS	GND
ETH_MDIO	PA2	VDD	V3V3
	PA3	PA14	SWCLK
GND	VSS	PA15	
V3V3	VDD	PC10	UART_TX
TFT_CS	PA4	PC11	UART_RX
TFT_SCK	PA5	PC12	
TFT_MISO	PA6	PD0	
ETH_CRS_DV	PA7	PD1	
ETH_RXD0	PC4	PD2	
ETH_RXD1	PC5	PD3	
TFT_DC	PB0	PD4	
TFT_RESET	PB1	PD5	
	PB2	VSS	GND
	PF11	VDD	V3V3
	PF12	PD6	
GND	VSS	PD7	
V3V3	VDD	PG9	
	PF13	PG10	
	PF14	PG11	ETH_TXEN
	PF15	PG12	
	PG0	PG13	ETH_TXD0
	PG1	PG14	ETH_TXD1
	PE7	VSS	GND
	PE8	VDD	V3V3
	PE9	PG15	
GND	VSS	PB3	SWO
V3V3	VDD	PB4	
	PE10	PB5	TFT_MOSI
	PE11	PB6	I2C_SCL
	PE12	PB7	I2C_SDA
	PE13	BOOT0	BOOT0
	PE14	PB8	CAN_RX
	PE15	PB9	CAN_TX
	PB10	PE0	
	PB11	PE1	
VCAP1	VCAP_1	PDR_ON	V3V3
V3V3	VDD	VDD	V3V3

J2 SWD

V3V3	1	V3V3	
SWDIO	2	SWDIO	
GND	3	GND	
SWCLK	4	SWCLK	
NRST	5	NRST	
SWO	6	SWO	

J1 POWER

V3V3	1	V3V3	
GND	2	GND	

R1 10k

BOOT0 — GND

C1 100nF

NRST — GND

02 Power and clock

Y1 ASE-8.000MHZ-LC-T

V3V3	1	OE	OUT 3 — HSE
GND	2	GND	VDD 4 — V3V3

R2 0R

V3V3 — VDDA

C2 2.2uF

VCAP1 — GND

C3 2.2uF

VCAP2 — GND

C4 10uF

V3V3 — GND

C5 10nF

V3V3 — GND

C6 1uF

VDDA — GND

C7 100nF

VDDA — GND

C10 100nF

V3V3 — GND

C11 100nF

V3V3 — GND

C12 100nF

V3V3 — GND

C13 100nF

V3V3 — GND

C14 100nF

V3V3 — GND

C15 100nF

V3V3 — GND

C16 100nF

V3V3 — GND

C17 100nF

V3V3 — GND

C18 100nF

V3V3 — GND

C19 100nF

V3V3 — GND

C20 100nF

V3V3 — GND

C21 100nF

V3V3 — GND

03 Serial interfaces

J3 UART

UART_TX	1	UART_TX	
UART_RX	2	UART_RX	
GND	3	GND	

U3 MAX3232ESE+

PUMP1P	1	C1P	R2OUT	9
PUMPVP	2	VP	T2IN	10 — V3V3
PUMP1N	3	C1N	T1IN	11 — RS232_TX
PUMP2P	4	C2P	R1OUT	12 — RS232_RX
PUMP2N	5	C2N	R1IN	13 — RS232_RX_LINE
PUMPVM	6	VM	T1OUT	14 — RS232_TX_LINE
	7	T2OUT	GND	15 — GND
GND	8	R2IN	VCC	16 — V3V3

J5 RS232

RS232_TX_LINE	1	RS232_TX_LINE	
RS232_RX_LINE	2	RS232_RX_LINE	
GND	3	GND	

C31 100nF

PUMP1P — PUMP1N

C32 100nF

PUMP2P — PUMP2N

C33 100nF

PUMPVP — GND

C34 100nF

PUMPVM — GND

C35 100nF

V3V3 — GND

U4 THVD1450DR

RS485_RX	1	RO	GND	5 — GND
RS485_DIR	2	RE_N	A	6 — RS485_A
RS485_DIR	3	DE	B	7 — RS485_B
RS485_TX	4	DI	VCC	8 — V3V3

J6 RS485

RS485_A	1	RS485_A
RS485_B	2	RS485_B
GND	3	GND

R5 10k

RS485_DIR — GND

C36 100nF

V3V3 — GND

JP1 RS485 TERM

RS485_A	1	RS485_A
RS485_TERM	2	RS485_TERM

R6 120R

RS485_TERM — RS485_B

04 CAN and Ethernet

U5 SN65HVD230DR

CAN_TX	1	D	VREF	5
GND	2	GND	CANL	6 — CAN_L
V3V3	3	VCC	CANH	7 — CAN_H
CAN_RX	4	R	RS	8 — GND

J7 CAN

CAN_H	1	CAN_H
CAN_L	2	CAN_L
GND	3	GND

C37 100nF

V3V3 — GND

R7 10k

CAN_TX — V3V3

JP2 CAN TERM

CAN_H	1	CAN_H
CAN_TERM	2	CAN_TERM

R8 120R

CAN_TERM — CAN_L

J8 RMII

V3V3	1	V3V3	
GND	2	GND	
ETH_REFCLK	3	ETH_REFCLK	
ETH_MDIO	4	ETH_MDIO	
ETH_MDC	5	ETH_MDC	
ETH_CRS_DV	6	ETH_CRS_DV	
ETH_RXD0	7	ETH_RXD0	
ETH_RXD1	8	ETH_RXD1	
ETH_TXEN	9	ETH_TXEN	
ETH_TXD0	10	ETH_TXD0	
ETH_TXD1	11	ETH_TXD1	
PHY_RESET	12	PHY_RESET	

R9 2.2k

ETH_MDIO — V3V3

R10 10k

PHY_RESET — V3V3

05 Display, EEPROM and USB

J4 TFT

V3V3	1	V3V3	
GND	2	GND	
TFT_SCK	3	TFT_SCK	
TFT_MOSI	4	TFT_MOSI	
TFT_MISO	5	TFT_MISO	
TFT_CS	6	TFT_CS	
TFT_DC	7	TFT_DC	
TFT_RESET	8	TFT_RESET	

U2 24LC256-I/SN

GND	1	A0	SDA	5 — I2C_SDA
GND	2	A1	SCL	6 — I2C_SCL
GND	3	A2	WP	7 — GND
GND	4	VSS	VCC	8 — V3V3

R3 4.7k

I2C_SDA — V3V3

R4 4.7k

I2C_SCL — V3V3

C30 100nF

V3V3 — GND

J9 USB FS

USB_VBUS	1	USB_VBUS	
USB_DM	2	USB_DM	
USB_DP	3	USB_DP	
GND	4	GND	

R11 33k

USB_VBUS — USB_VBUS_SENSE

R12 47k

USB_VBUS_SENSE — GND

C38 10uF

V3V3 — GND